

Splunk Mastery Lab User Guide

Step-by-step instructions for running Splunk, using the dashboards, launching the Agentic SOC Analyst, reviewing reports, and enabling optional home-network telemetry.

Local URL: `http://localhost:8000`

Default login: admin / SplunkLab!2026

What this guide is for

This guide assumes Docker Desktop is installed and that the lab files live at `C:\Users\Joela\Documents\Codex\2026-06-18\can-you-please-build-and-design\outputs\splunk-mastery-lab`.

1. Quick Start

Use this section when you simply want to start the lab and open Splunk.

Step 1 - Open Docker Desktop

Start Docker Desktop from Windows. Wait until Docker Desktop says the engine is running.

Step 2 - Open PowerShell

```
cd C:\Users\Joela\Documents\Codex\2026-06-18\can-you-please-build-and-design\outputs\splunk-mastery-lab
docker compose up -d
```

Step 3 - Open Splunk

In a browser, open: <http://localhost:8000>

```
username: admin
password: SplunkLab!2026
```

Step 4 - Confirm the App

- Open the Splunk app menu.
- Choose **Splunk Mastery Lab**.
- Open the dashboards from the Dashboards navigation collection.

Good sign

If Splunk Web loads and the app is visible, the lab is ready for dashboard review and agent triage.

2. Clean Rebuild

Use this only when you want to reset the lab data and rebuild from a clean container.

```
cd C:\Users\Joela\Documents\Codex\2026-06-18\can-you-please-build-and-design\outputs\splunk-mastery-lab
docker compose down -v
docker compose up -d
.\scripts\Install-LabIntoContainer.ps1
```

Important

A clean rebuild deletes the current Splunk container state and reloads the generated lab data. Do not use it if you want to preserve current investigations.

3. What Is Inside the Lab

Component	Purpose
Splunk Enterprise container	Runs local Splunk Web, management API, HEC, dashboards, and searches.

Component	Purpose
Splunk Mastery Lab app	Contains indexes, sourcetypes, lookups, dashboards, macros, and saved searches.
Generated telemetry	Auth, web, endpoint, cloud, and network JSONL data with seeded incident storylines.
Agentic SOC Analyst	External Python agent that queries Splunk and writes triage reports.
Home-network scripts	Optional Windows connection telemetry, pktmon packet capture, and router syslog ingestion.

Lab Indexes

Index	Data
lab_auth	Authentication, MFA, user, source IP, source country, and identity signals.
lab_web	HTTP/API requests, status codes, latency, user agents, and web attack types.
lab_endpoint	Endpoint process, command line, parent process, and network egress events.
lab_cloud	AWS-style CloudTrail actions, identity principals, regions, source IPs, and errors.
lab_network	Network flow, ports, bytes, packets, signatures, and router/live telemetry.

4. Using Splunk Search

Start with these validation searches in Splunk Search.

Confirm data exists

```
| eventcount summarize=false index=lab_auth index=lab_web index=lab_endpoint index=lab_cloud index=lab_network
| table index count
| sort index
```

Search all lab events

```
index=lab_auth OR index=lab_web OR index=lab_endpoint OR index=lab_cloud OR index=lab_network
| stats count by index sourcetype
| sort index sourcetype
```

Find high-severity activity

```
(index=lab_auth OR index=lab_web OR index=lab_endpoint OR index=lab_cloud OR index=lab_network)
severity IN ("high","critical")
| stats count values(sourcetype) as sourcetypes by index severity
| sort - count
```

5. Dashboards

Open the **Splunk Mastery Lab** app, then use the dashboards collection.

Dashboard	Use it for
SOC Overview	Executive view of signal volume, high-severity events, affected users, and prioritized entities.
Auth and Identity Threats	Brute force, password spray, impossible travel, MFA outcomes, and authentication trends.
Web and API Security	HTTP status trends, web attack distribution, risky clients, and p95 service response time.
Endpoint Detection	Suspicious process chains, living-off-the-land tools, and endpoint egress.
Cloud Security	CloudTrail actions, API errors, privilege escalation, and unusual source countries.
Dashboard habit Set the time picker to Last 24 hours for live-style triage, or Last 7 days when you want broader historical context.	

6. Running the Agentic SOC Analyst

The agent runs from PowerShell and connects to Splunk over **https://localhost:8089**. It is read-only: it searches, summarizes, ranks, and writes reports.

One-command visual report

```
cd C:\Users\Joela\Documents\Codex\2026-06-18\can-you-please-build-and-design
cd outputs\splunk-mastery-lab
.\scripts\Run-AgentDashboard.ps1
```

This command creates and opens **reports\latest_triage.html**.

Full manual command

```
python -m agent.run_triage --mode all --earliest -24h --latest now `
--report reports\latest_triage.md `
--html-report reports\latest_triage.html `
--json-output reports\latest_triage.json
```

Focused modes

```
python -m agent.run_triage --mode endpoint --earliest -24h
python -m agent.run_triage --mode identity --earliest -7d
python -m agent.run_triage --mode cloud --earliest -7d
python -m agent.run_triage --mode web --earliest -24h
python -m agent.run_triage --mode network --earliest -24h
```

7. Reading the Agent Report

Report area	What it means
Severity cards	Counts of critical, high, medium, and low finding groups.
Finding domains	Which telemetry domains produced suspicious activity.
Priority queue	Highest-priority finding groups first.

Report area	What it means
Evidence table	The actual Splunk rows returned by the detection SPL.
SPL used by the agent	Expandable search logic so reviewers can see exactly how the finding was generated.
Recommendations	Human-in-the-loop next steps for investigation.
Safety model The agent does not block IPs, kill processes, quarantine files, or modify endpoints. It is built for diagnosis and investigation guidance.	

8. Optional Home-Network Telemetry

The lab can ingest live telemetry from your own PC and router. Only monitor networks and devices you own or are authorized to monitor.

Windows connection telemetry

```
.\scripts\Watch-NetConnections.ps1 -Seconds 900 -IntervalSeconds 5
```

This writes JSONL connection metadata that Splunk can ingest as **splunk_lab:windows_netconn:json**.

Packet capture

Run PowerShell as Administrator, then:

```
.\scripts\Start-PktmonCapture.ps1 -Seconds 120
```

This creates a PCAPNG file under **captures** for Wireshark-style analysis.

Router syslog

If your router supports syslog forwarding, send logs to your PC on UDP port **5514**. Search them in Splunk with:

```
index=lab_network sourcetype=splunk_lab:router:syslog
| stats count by host source
```

9. Troubleshooting

Symptom	Fix
Splunk page does not load	Open Docker Desktop, then run <code>docker compose up -d</code> from the lab folder.
Login fails	Use <code>admin / SplunkLab!2026</code> unless you changed <code>.env</code> .
App is missing	Run <code>.\scripts\Install-LabIntoContainer.ps1</code> after Splunk Web is reachable.
Agent cannot connect	Confirm Splunk is reachable at <code>http://localhost:8000</code> and management API at <code>https://localhost:8089</code> .
Docker shows unhealthy	Check Splunk Web first. A healthcheck label can be wrong even when Splunk Web and API are working.

Symptom	Fix
No report opens	Run <code>python -m agent.run_triage</code> manually and inspect any PowerShell error output.
Need a clean reset	Run <code>docker compose down -v</code> , <code>docker compose up -d</code> , then <code>.\scripts\Install-LabIntoContainer.ps1</code> .

10. GitHub Presentation Tips

- Keep the README screenshots or report screenshots visible near the top of the repository.
- Explain that the lab includes Splunk administration, data onboarding, SPL analytics, dashboards, and agentic triage.
- Mention the agent is defensive and human-in-the-loop.
- Commit the source code, dashboards, docs, and generated sample data.
- Do not commit sensitive packet captures from your real network.

Suggested demo flow

- Start Splunk with Docker.
- Open SOC Overview and explain the indexes.
- Run the Agentic SOC Analyst.
- Open the HTML report.
- Pivot from one finding back into Splunk Search.
- Show the SPL playbook and app configuration files in GitHub.

11. Key Commands Reference

```
# Start existing lab
docker compose up -d

# Clean rebuild
docker compose down -v
docker compose up -d
.\scripts\Install-LabIntoContainer.ps1

# Run visual agent report
.\scripts\Run-AgentDashboard.ps1

# Run focused endpoint triage
python -m agent.run_triage --mode endpoint --earliest -24h

# Stop lab
docker compose down
```